

Graphic Method - 2D problem

Descent Method $X_k = X_k + \alpha S_k$, S_k search direction, α - step length

Rate of convergence $\lim_{k \rightarrow \infty} \frac{\|X_{k+1} - X^*\|}{\|X_k - X^*\|} = \rho$
 $\rho = 1, 1 < \rho < 2$ linear, superlinear, quadratic

Termination $\|X_{k+1} - X_k\| \leq \epsilon$ or $|f(X_{k+1}) - f(X_k)| \leq \epsilon$ or $\left| \frac{f(X_{k+1}) - f(X_k)}{f(X_k)} \right| \leq \epsilon$ or $\|\nabla f(X_{k+1})\| \leq \epsilon$

Directional derivative $\frac{\partial f(X_k)}{\partial S} = [\nabla f(X_k)]^T S$, $S_0 = [\cos \alpha_1, \cos \alpha_2, \dots, \cos \alpha_n]^T$

Taylor Expansion $f(x) = f(x_k) + f'(x_k) \cdot (x - x_k) + \frac{1}{2} f''(x_k) \cdot (x - x_k)^2 + \dots + R_n$

Quadratic ~ $f(X) \approx f(X_k) + [\nabla f(X_k)]^T (X - X_k) + \frac{1}{2} (X - X_k)^T \nabla^2 f(X_k) (X - X_k)$

convex set/function $x, y \in S$ $\alpha x + (1-\alpha)y \in S$ / $f[\alpha x + (1-\alpha)y] \leq \alpha f(x) + (1-\alpha)f(y)$

- 阶必要: f - 阶可微, $\nabla f(x^*) = 0 \Leftrightarrow x^*$ 驻点; local min $\Rightarrow \nabla f(x^*) = 0$

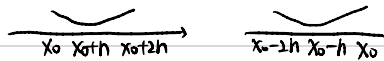
= 阶必要: x^* local min + ∇f 连续 in Nof $x^* \Rightarrow \nabla f(x^*) = 0$ & $\nabla^2 f(x^*)$ PSD.

= 阶充分: ∇f 连续 in Nof x^* , $\nabla f(x^*) = 0$, $\nabla^2 f(x^*)$ p.d. $\Rightarrow x^*$ strict local min

f convex local min $\Leftrightarrow$ global min, + differential, stationary = global

Chapter 3 Linear Search Method

determine the initial interval: $x_0, h=1.0$



Golden Section Method $\tau = \frac{b_k - a_k}{b_{k-1} - a_{k-1}} = 0.618$

$$x_1 = a + 0.382(b-a), x_2 = a + 0.618(b-a)$$

Fibonacci Search Method $0 \ 1 \ 2 \ 3 \ \dots$

$$\tau_1 = \frac{L_1}{L_1} = \frac{F_{n-1}}{F_n} \dots L_n = \frac{F_1}{F_n} L_1 \quad 1 \ 1 \ 2 \ 3 \ \dots$$

STEP 1: $L_n = (F/F_n) L_1 = (b-a)/F_n \leq \epsilon \Rightarrow F_n \geq (b-a)/\epsilon \Rightarrow n$

STEP 2: $x_1 = a + (1 - \tau_k)(b-a)$, $x_2 = a + \tau_k(b-a)$

$f(x_1) < f(x_2)$, $[a, x_2]$; $f(x_1) > f(x_2)$, $[x_1, b]$

Powell's QIM (二次插值) $[a, b]$ have c

$a < c < b$, $f_a > f_c < f_b$, $x_1 = a, x_2 = c, x_3 = b$.

STEP 1: $x_p = -\frac{\alpha_1}{2\alpha_2} = \frac{1}{2} \frac{(x_2^2 - x_3^2)f_1 + (x_3^2 - x_1^2)f_2 + (x_1^2 - x_2^2)f_3}{(x_2 - x_3)f_1 + (x_3 - x_1)f_2 + (x_1 - x_2)f_3}$

$$\alpha_2 = -\frac{(x_2 - x_3)f_1 + (x_3 - x_1)f_2 + (x_1 - x_2)f_3}{(x_2 - x_3)(x_3 - x_1)(x_1 - x_2)} > 0$$

STEP 2: i) $x_p \leq x_2, f_p \leq f_2$, new IU is $[x_1, x_2]$ having x_p ;

ii) $x_p \leq x_2, f_p > f_2$, new IU is $[x_p, x_3]$ including x_2 ;

iii) $x_p > x_2, f_p \leq f_2$, new IU is $[x_2, x_3]$ holding x_p ;

iv) $x_p > x_2, f_p > f_2$, new IU is $[x_1, x_p]$ containing x_2 .

STEP 3: $|x_2 - x_p| \leq \epsilon$, $f^* = \min\{f_2, f_p\}$, $x^* = f^{-1}(f^*)$

Davidon's CIM (三次插值) if f' is easy to evaluate.

1: $f_0 = f(x_0)$, $G_0 = f'(x_0)$, check $G_0 < 0$, choose $k(=2)$ and $f_e \Rightarrow \alpha$

2. evaluate $f_\alpha = f(x_0 + \alpha)$, G_α , $\alpha = \min\left\{k, \frac{-2(f_0 - f_e)}{G_0}\right\}$
if $G_\alpha > 0$, $f_\alpha > f_0$, go ahead, else $\alpha = 2\alpha$,

3. Interpolate: $Z = G_0 + G_\alpha + 3(f_0 - f_\alpha)/\alpha$, $w = (Z - G_0 G_\alpha)^{1/2}$

$$\lambda_m / \alpha = (G_0 + Z + w) / (G_0 + G_\alpha + 2Z) = (Z + w - G_0) / (G_\alpha + 2w - G_0)$$

$[x_0, x_0 + \lambda_m]$ OR $[x_0 + \lambda_m, x_0 + \alpha]$ IF $f'(x_0 + \lambda_m) \geq 0$ or $f(x_0 + \lambda_m) > f(x_0)$
OR $f'(x_0 + \lambda_m) < 0$

4. $\lambda_m \leq \epsilon$ or length of the interval $< \epsilon$

Chapter 4 Unconstrained Optimization Methods.

Steepest Descent Method (SDM)

1. Initial X_0 and convergence accuracy $\epsilon > 0$, $k=0$

2. $S_k = -\nabla f(X_k)$, $f(X_k + \alpha_k S_k) = \min f(X_k + \alpha S_k)$

$$X_{k+1} = X_k + \alpha_k S_k$$

3. STOP until $\|\nabla f(X_{k+1})\| \leq \epsilon$, $X^* = X_{k+1}$

Newton's Method

$S_k = -[\nabla^2 f(X_k)]^{-1} \nabla f(X_k)$, others same as SDM

Quasi-Newton's Method (has the superlinear convergence rate)

1. $X_0, H_0 = I$, $\epsilon > 0$,

2. $k=0$, $g_k = \nabla f(X_k)$,

3. $p_k = -H_k g_k$, 1-D search: $X_{k+1} = X_k + \alpha_k p_k$

4. update: $H_{k+1} = H_k + \Delta H_k$, $H_{k+1} = H_k + \frac{S_k S_k^T}{S_k^T Y_k} - \frac{H_k Y_k (H_k Y_k)^T}{Y_k^T H_k Y_k}$ DFP

$$S_k = X_{k+1} - X_k, Y_k = g_{k+1} - g_k$$

5. Until $\|\nabla f(X_{k+1})\| \leq \epsilon$, $X^* = X_{k+1}$

- 阶校正: $H_{k+1} = H_k + \frac{(S_k - H_k Y_k)(S_k - H_k Y_k)^T}{(S_k - H_k Y_k)^T Y_k}$ = 阶 $\uparrow$

BFGS

$$H_{k+1} = H_k + (1 + \frac{Y_k^T H_k Y_k}{S_k^T Y_k}) \frac{S_k S_k^T}{S_k^T Y_k} - \frac{S_k Y_k^T H_k + H_k Y_k S_k^T}{S_k^T Y_k}$$

Conjugate Gradient Method

conjugate Direction:

H 对称正定, 非零 $S_1, \dots, S_n$, $S_i^T H S_j = 0$ ($i \neq j$)

$\{S_1, \dots, S_n\}$ are called as the set of conjugate vectors (dire- of H)

$H = I$, $S_i^T S_j = 0$, 相互正交

properties: (1) S_i 是正定对称 H 共轭方向, S_i ($i=1, \dots, n$) 线性独立

(2) 正定对称阵 H 的不同共轭方向不超过 n .

(3) for 正定二次型函数, 从任意点出发, 沿着共轭方向至多 n 次迭代可达到 min.

1. Given X_0 , $\epsilon > 0$, $p_0 = -g_0 = -\nabla f(X_0)$, $G_0 = \nabla^2 f(X_0)$; Set $k=0$;

2. 1-D search: $f(X_k + \alpha_k p_k) = \min f(X_k + \alpha p_k) \Rightarrow \alpha_k$

3. $X_{k+1} = X_k + \alpha_k p_k \Rightarrow g_{k+1} = \nabla f(X_{k+1})$, $G_{k+1} = \nabla^2 f(X_{k+1})$

4. STOP: $\|\nabla f(X_{k+1})\| \leq \epsilon$, $X^* = X_{k+1}$

5. $p_{k+1} = -g_{k+1} + \beta_k p_k$, $\beta_k = \frac{g_{k+1}^T G_{k+1} p_k}{p_k^T G_{k+1} p_k}$

PRP

$$\beta_k = \frac{g_{k+1}^T (g_{k+1} - g_k)}{\|g_k\|^2}$$

FR

$$\beta_k = \frac{\|g_{k+1}\|^2}{\|g_k\|^2}$$

Problem	Characteristics	Features of Algorithm	Algorithms / Methods
Linear	linear functions of variables	Vortex Conversion	Simplex Method
Nonlinear	Unconstrained Optimization	One-dimensional Search	Golden Section Search Method, Fibonacci Search Method, Quadratic Interpolation Method, Cubic Interpolation Method.
		Use information of derivatives	Steepest Descent Method, Newton Method, Quasi-Newton Method, Conjugate Gradient Method.
		Don't use info.	Powell Method
	Constrained Optimization	Solve directly	Feasible Direction Method
		Solve indirectly	Penalty Function Method, SQP Method, etc.

Chapter 5 Linear Programming

General Form: $\min(\max) f(x) = C^T x$

$$\text{s.t. } Ax = b \quad m \times n$$

$$x_i \geq 0 \quad i=1, \dots, n$$

基本解: 只满足约束方程; **基本可行解**: 满足约束方程+非负约束

最优解: 取得最优函数值的基本可行解.

[A|b]初等行变换为单位阵, 通过选取不同 x_i 列作为1列, 可以得到不同系列的基本解系

The Simplex Method

目标是得到最优解, 方法是使所有检验数非负

$$\textcircled{1} \min f(x) = \sum c_i x_i + C_0 \Rightarrow \min f(x) = \sum c_i x_i + C_0$$

$$\text{先化标准型 } Ax \leq b \text{ Slack } Ax + \begin{bmatrix} x_{n+1} & 0 \\ 0 & x_{n+m} \end{bmatrix} = b$$

1列对应为基变量, 基变量检验数为0,

非基变量 $\sigma_j = c_j - \sum c_i a_{ij}$, C_j 为上方一行系数, C_i 为左方基变量列系数

SIMPLEX TABLE		x_1	x_2	$\dots$	x_n	x_{n+1}	x_{n+2}	$\dots$	x_{n+m}	b_i
Basic Variable	Coefficient	c_1	c_2	$\dots$	c_n	0	0	$\dots$	0	c_0
x_{n+1}	0	a_{11}	a_{12}	$\dots$	a_{1n}	1	0	$\dots$	0	b_1
x_{n+2}	0	a_{21}	a_{22}	$\dots$	a_{2n}	0	1	$\dots$	0	b_2
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	0	0	$\dots$	0	$\vdots$
x_{n+m}	0	a_{m1}	a_{m2}	$\dots$	a_{mn}	0	0	$\dots$	1	b_m
Judgment number σ_j		σ_1	σ_2	$\dots$	σ_n	0	0	$\dots$	0	$f(X)$

进基(列): $k = \min \{j | \sigma_j < 0\}$ / $6k = \min \{i | a_{ik} > 0\}$ 使 k 列成为基变量

出基(行): 在 k 列找 l 行 $\frac{b_l}{a_{lk}} = \min \left\{ \frac{b_i}{a_{ik}} \mid a_{ik} > 0 \quad (i=1, 2, \dots, m) \right\}$

Choose a_{lk} as the principal element (行变换将 l 列除 a_{lk}

外变为0, a_{lk} 变为1)

重写左列基变量 Coefficient, 重新计算 σ_j , $f(x) = C_0 + \sum C_j b_j$

重复上述过程, 直至检验数全部非负

$$\textcircled{2} \min f(x) = CX \quad \text{可能找不出一组基变量}$$

$$Ax = b$$

$$x_i \geq 0, \quad i=1, \dots, n$$

introduce the artificial variables

$$\min \phi(x) = x_{n+1} + \dots + x_{n+m} \quad \text{辅助问题}$$

$$Ax + \begin{bmatrix} x_{n+1} & 0 \\ 0 & x_{n+m} \end{bmatrix} = b$$

$$x_i \geq 0, \quad i=1, \dots, n, n+1, \dots, n+m$$

Simplex Method $\rightarrow x_{n+1} = \dots = x_{n+m} = 0$ (非基变量)

去除人工变量部分, 继续求解原问题

Basic Variables	Coefficients	x_1	x_2	x_3	x_4	x_5	b_i
		0	0	0	1	1	0
x_3	0	0	-3/4	1	3/4	-1/2	3/2
x_1	0	1	5/4	0	-3/4	1/2	1/2
σ_j		0	0	0	1	1	0



Basic Variables	Coefficients	x_1	x_2	x_3	b_i
		1	1	0	0
x_3	0	0	-3/4	1	3/2
x_1	1	1	5/4	0	1/2
σ_j		0	-1/4	0	1/2

For Functions with multiple variables:

Taylor's Theorem:

1) If f is once continuously differentiable in an open neighborhood of X^* and p is a non-zero vector, then:

$$f(X^* + p) = f(X^*) + \nabla f(X^* + \lambda p)^T p \quad \text{for } \lambda \in (0, 1)$$

2) If f is twice continuously differentiable, then:

$$f(X^* + p) = f(X^*) + \nabla f(X^*)^T p + \frac{1}{2} p^T \nabla^2 f(X^* + \lambda p)^T p$$

Chapter 6. Constrained Optimization Methods

- 一般形式 $\min f(x)$

$$\text{s.t. } g_u(x) \leq 0 \quad (u=1, 2, \dots, p)$$

$$h_v(x) = 0 \quad (v=1, 2, \dots, m)$$

effective constraints: 在约束的约束边界上

$$\text{有效约束集合 } I_k = \{u \mid g_u(x) = 0, u=1, \dots, p\}$$

$$\textcircled{1} \text{等式约束极值条件 } L(x, \lambda) = f(x) + \sum \lambda_v h_v(x)$$

等式约束在 X 点取得极值必要条件:

$$\nabla f(x) + \sum \lambda_v \nabla h_v(x) = 0, \quad \lambda_v \text{ are not all zeros. (1)}$$

$$\textcircled{2} \text{不等式约束极值条件 } \min f(x) \quad \Leftrightarrow \quad \min f(x) \quad \text{s.t. } g_u(x) \leq 0 \quad g_u(x) + x_{n+u}^2 = 0$$

不等式约束在 X 点取得极值必要条件:

$$\nabla f(x) + \sum \lambda_i \nabla g_i(x) = 0 \quad \lambda_i \geq 0 \quad (i \in I_k) \quad (2)$$

表明在不等式问题的极值点处, 目标函数的负梯度等于有效约束梯度的非负线性组合。

K-T condition: (1) 式, (2) 式; 约束问题达到极值必要条件

Feasible Direction Method

用于解决不等式约束问题, $\min f(x) \quad \text{s.t. } g_u(x) \leq 0 \quad (u=1, \dots, p)$

Optimal Descent Feasible Direction: X_k 所有下降可行方向, 下降速度最大

核心: 找到最优下降可行方向, 一维搜索最优步长

STEP 1: Given X_0, ε, δ , set $k=0$

STEP 2: identify the **effective constraints** of point X_k :

$$I_k(X_k, \delta) = \{u \mid -\delta \leq g_u(X_k) \leq \delta\} \quad (u=1, 2, \dots, p)$$

STEP 3: if $I_k = \text{null}$, X_k 位于可行域内, if $\|\nabla f(X_k)\| \leq \varepsilon$, terminated

$$X^* = X_k, f^* = f(X_k), \text{ otherwise } S_k = -\nabla f(X_k), \text{ to STEP 6}$$

STEP 4: $I_k \neq \text{null}$, if X_k satisfy the K-T condition

$$\nabla f(X_k) + \sum \lambda_u \nabla g_u(X_k) = 0, \quad \lambda_u \geq 0, \text{ terminated.}$$

STEP 5: seek ODFD $\min [\nabla f(X_k)]^T S$

$$\text{s.t. } [\nabla g_u(X_k)]^T S \leq 0 \quad (u \in I_k)$$

$$S_k = S^* \quad -1 \leq s_j \leq 1$$

STEP 6: along S_k , 1-D Search $X_{k+1} = X_k + \alpha S_k$

一维搜索出的 $\alpha^* \rightarrow X_{k+1}$ 可能不在可行域内, 需作判别